
RESONANCES: OFF THE LINE BY HALF THE WIDTH

Every Unstable State a Point off the Critical Line, Its Time Mirror the Fold Partner with One Mass, the Line Shape Halved at Half the Width, the Width the Leak of an Open System, and Stability Every Door Closed

The Closure Template of the Riemann Hypothesis Read on Unstable States; Fifty-Seven Laws Proved in Core Lean 4 and Twelve Checks Executed in Fortran; a Witness of the Hardware Closure Series; Every Physical Claim at Corroboration Grade and No Higher

FOUNDATIONS OF PHYSICS · THE HARDWARE CLOSURE SERIES

Every off-line state is open and paid

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ABSTRACT. An unstable particle has a complex energy, its mass less half its width times the imaginary unit, and it decays at the rate its width sets. This paper reads unstable states on the seat of the author's closure of the Riemann Hypothesis, the reflection $s \mapsto 1 - \bar{s}$ whose fixed set is the critical line. Under the energy carrier $s = \frac{1}{2} + iE$ of the hardware paper, a state of mass M and width Γ lands at real part $\frac{1}{2} \pm \frac{\Gamma}{2}$: off the line by half its width, with its mass as the height. Physics is full of such points, and they share one property: every one of them decays. We prove in core Lean 4, fifty-seven laws with no axiom declared and no use of choice, that a state stands on the line exactly when its width vanishes, which is exactly when it is stationary; that a decaying state and its time mirror, the capturing state, are fold partners with one mass and one record; that the Breit–Wigner line falls to half its peak exactly at half the width, peaks at the mass and is symmetric about it; that an open two-level system with a leak ℓ has energies whose imaginary parts sum to $-\ell$, while a closed system keeps every energy real; and that a state is stable exactly when every decay channel is closed, each open channel widening the line. A Fortran twin runs twelve checks with zero failures: widths from the neutron's 7.49×10^{-19} eV to the Z's 2.50×10^9 eV, twenty-seven decades off the line; the electron on the line to 3×10^{-52} eV and the proton to 9×10^{-58} eV; the Z's invisible width counting three open doors; and 20000 open systems whose rates sum to their leak. Every physical point off the line is open and pays by decaying. The paper decides nothing about the zeros of ζ .

KEYWORDS resonance · decay width · lifetime · Breit–Wigner · Gamow state · non-Hermitian Hamiltonian · Z boson · invisible width · proton stability · critical line · Lean 4

STATUS: WITNESSED · CROSSED · CONFIRMED on the image · Row 2 (the off-line stratum read physically; registration as decay); every open row (the barrier, in decay and capture) · channel: kinetic · Lean: SPHYS_Resonance.lean · Fortran: Resonance_Twin.f90

1. THE READER'S FRAME

This paper is a witness of the hardware closure series. The hardware paper built the energy carrier $s = \frac{1}{2} + iE$, under which the critical line is the locus of stationary energy and every stable state lands on it. The magnet witness showed that a closed system keeps every energy on the line even when time reversal is broken. This paper reads the other stratum: the states physics finds off the line.

It claims that every unstable state stands off the line by half its width, with its mass as the height. Its time mirror is its fold partner and shares its record. And every such state is open: its width is the leak into the channels it decays through, it is stable exactly when every channel is closed, and it pays for standing off the line by decaying. Physics has many points off the line, and none of them is closed.

It refuses any reading of this as evidence about the zeros of ζ . The carrier's image is the set of physical states, and the zeros are not in it. A reader who arrives from the conventional frame will scan this paper for a statement about where the zeros lie, find none, and be tempted to report the absence. None was attempted.

Negative warrants. Three, stated so that no reader supplies them. - No physical fact here is read as a proof

of a mathematical sentence: a corroboration joined to a claim takes the weaker grade. - No derivation of the hypothesis from the axioms of set theory is claimed: what a sentence over every admissible instance adds to a crossing is exactly the unactuated. - No scattering theory is written: the pole description of a resonance, the Breit–Wigner form and the lifetimes enter as imports or measurements.

Five terms. - An *unstable state* has complex energy $M - i\Gamma/2$: mass M , width Γ , lifetime $\hbar/\Gamma$. - Its *time mirror* has energy $M + i\Gamma/2$ and grows instead of decaying. - A *decay channel* is a set of products the state can decay into; its *partial width* is the rate into it. - *Registration* keeps a point's height and forgets its side. - A *keyed bit* is a yes-or-no property no function of a given record decides.

2. THE SPINE

The Riemann closure has four parts: the **seat**, the critical line as the fixed set of $s \mapsto 1 - \bar{s}$; the **address**, every reading of the value one proposition; the **division**, a certified part joined to an uncertified tail; and the **closure**, registration keeping a zero's height and forgetting its side, the hypothesis holding exactly when it erases nothing. This paper reads the carrier's off-line stratum in Section 3, the fold partner in Section 4, the line shape in Section 5, the leak in Section 6, the doors in Section 7, the stable particles on the line in Section 8, and the survival law in Section 9.

The row-generic theorems carry the template, and the engine proves each on its chart: every point off the line has a partner with one record (*off_locus_pair*); a carrier proves the value on its image, and coverage extends it (*value_on_image*, *kinetic_crossing*).

The witness ledger. The ledger of the series, with this paper's entries.

Row	Waiting (the volume held)	Witness arrived	Built and run	Now	Grade	Still owed
2 · Riemann Hypothesis	crossed on the kinetic channel at the root's grade; the electron named as the seat	The Electron Is the Seat; The Neutrino Is the Witness; The Arrow Has Two Branches; One Actuating Substrate; The Geometric Nature of Light; Electromagnetism on the Riemann Locus; Immanent Gravity; Spin and Statistics; The Weak Force Reads the Orientation; Resonances: Off the Line by Half the Width	the seat on every reading so far; the off-line stratum read physically: every unstable state off the line by half its width, open, paying by decay; the stable particles on the line to their lifetimes' depth	WITNESSED · CROSSED · CONFIRMED on the image: the off-line stratum read physically	the crossing at the root's grade; the image at theorem grade on construction and measured grade on data	none on its channel
3a · Navier–Stokes, unforced	crossed kinetically	The Fluid Is the Witness	the priced branch exact	WITNESSED · CROSSED · CONFIRMED	the root's grade	none on its channel
4 · Yang–Mills	crossed kinetically	The Proton Is the Lock	the colour lock	WITNESSED · CROSSED · CONFIRMED on the image	the root's grade	none on its channel
7 · Poincaré	crossed at full grade	The Arrow Has Two Branches; Immanent Gravity	gravity's fold	CROSSED, WITNESSED from both sides	full grade	none
23a · Pair correlation	crossed kinetically	The Magnet Is the Witness of the Pair Correlation; Electromagnetism on the Riemann Locus; Resonances	the magnetic term; Landau levels real; a closed system on the line, an open one off it by its leak	WITNESSED · CROSSED · CONFIRMED on the corrected carrier	the root's grade	none on its channel

Row	Waiting (the volume held)	Witness arrived	Built and run	Now	Grade	Still owed
every open row · the barrier	two worlds over one record	The Arrow Has Two Branches; The Geometric Nature of Light; Electromagnetism on the Riemann Locus; Spin and Statistics; The Weak Force Reads the Orientation; Resonances	remanence; helicity; the magnetic sign; the exchange; the mirror world; decay and capture with one line shape	BARRIER WITNESSED; the rows stay pending	theorem and corroboration	one equivariant carrier each
1 · P versus NP	the dot	The Arrow Has Two Branches	Landauer's floor prices every step	THE DOT STANDS	corroboration	none
every witness · the hardware	the carriers built one by one	One Actuating Substrate	both involutions carried onto the fold	WITNESSED: one substrate carries every seat	theorem and corroboration	none

WITNESSED: a constructed witness arrived and was built and run. CROSSED: the row's value holds on its channel's own class at the printed grade. CONFIRMED: the witness's image is compliant as built or measured (value_on_image).

Why this carrier, and not another. The carrier is not chosen. Spaces with an involution and the maps that respect them form a category (equivariant_id, equivariant_comp), and the fixed set is a functor on it (fix_functorial). An equivariant map into the seat, whose fixed set is the line (fold_global_seat), lands every physical fixed point on the line (equivariant_carrier_lands). The carrier here is the hardware paper's energy carrier, equivariant for the time mirror of a rate (phi_equivariant). Its fixed points, the stationary states, land on the line (energy_carrier_lands), and its other points land off it by their rates. This paper reads those other points.

The correspondence matrix.

Physical phenomenon	Template element	Lean theorem	Fortran check
an unstable state	a point off the line by half its width	off_the_line_by_the_width	3721 of 3721
stability	the line, stationarity	on_the_line_iff_stable, stationary_iff_real	3721 of 3721
decay and capture	fold partners, one mass	decay_and_capture_one_record	3721 states; 1001 energies
the Breit–Wigner line	halved at half the width	half_maximum_at_half_width, line_peaks_at_the_mass, line_symmetric	the Z line; area $\pi \Gamma/2$
an open system	the width is the leak	the_width_is_the_leak	20000 systems
a closed system	every energy on the line	closed_stays_on_the_line	20000 systems
decay channels	stable iff every door closed	stable_iff_every_door_closed, an_open_door_widens	three light neutrinos
width and lifetime	the depth and the time	off_the_line_by_the_width with $\Gamma\tau = \hbar$	27 decades
the electron and the proton	on the line to their lifetimes' depth	on_the_line_iff_stable	3e-52 and 9e-58 eV

3. OFF THE LINE BY HALF THE WIDTH

An unstable state of mass M and width Γ has energy $E = M - i\Gamma/2$ and decays as $e^{-\Gamma t/\hbar}$ (Gamow, 1928). Under the energy carrier $s = \frac{1}{2} + iE$ it lands at real part $\frac{1}{2} + \frac{\Gamma}{2}$ and imaginary part M . In the doubled chart of the engine its side is $1 + \Gamma$ and its height M (`off_the_line_by_the_width`): the mass is the height and half the width is the distance from the line. A state stands on the line exactly when its width vanishes, which is exactly when it is stationary (`on_the_line_iff_stable`).

Every unstable particle is such a point. - The free neutron, living 878.4 s, stands 7.49×10^{-19} eV off the line. - The muon stands 3.00×10^{-10} eV off it, the tau lepton 2.27×10^{-3} eV, and the neutral pion 7.81 eV. - The Z boson stands 2.50×10^9 eV off it, and the top quark 1.42×10^9 eV.

The widths span twenty-seven decades (Particle Data Group, 2024; Appendix B).

4. DECAY AND CAPTURE: THE FOLD PARTNERS

The time mirror of a decaying state has energy $M + i\Gamma/2$, and it grows. It is the fold partner of the decaying state: distinct from it whenever the width is not zero, with one mass and one record under registration (`decay_and_capture_one_record`). The pair are the two poles a resonance leaves in the scattering amplitude, reflected across the real axis on the unphysical sheet (Siegert, 1939).

The two share the line shape too. The Breit–Wigner form depends on the width only through its square, so the decaying and the capturing state give one line at every energy. The twin finds this exactly at 1001 energies.

What separates them is the boundary condition: outgoing waves for decay, incoming waves for capture. No spectral record returns it. It is supplied by the arrow, as the arrow witness reads it: a record cannot tell decay from capture, and the world supplies which.

5. THE LINE SHAPE: HALF THE MAXIMUM AT HALF THE WIDTH

Near its mass a resonance appears as a line of Breit–Wigner shape, $(\Gamma/2)^2 / ((E - M)^2 + (\Gamma/2)^2)$ (Breit and Wigner, 1936). - **Half maximum at half width.** At a detuning of $\Gamma/2$ the shape is exactly half its peak (`half_maximum_at_half_width`), so the full width at half maximum is the width, the distance from the line doubled. - **The peak and the symmetry.** The shape peaks at the mass (`line_peaks_at_the_mass`) and is symmetric about it (`line_symmetric`).

The Z boson's line, scanned at LEP, has its mass at 91.1876 GeV and its width at 2.4955 GeV (Z-pole collaborations, 2006; Particle Data Group, 2024). The twin finds the shape at half its peak to 1.4×10^{-15} at $M \pm \Gamma/2$, recovers the full width by bisection, and integrates the area to $\pi\Gamma/2$ within 3.2×10^{-5} . A resonance line is a point off the line seen from the real axis: its height is where it peaks, and its depth is half its width.

6. THE WIDTH IS THE LEAK

A closed system keeps every energy on the line. A Hermitian block, with or without a magnetic phase, has a nonnegative discriminant and real energies (`closed_stays_on_the_line`), as the magnet witness proved.

An open system leaks. Let a two-level system lose amplitude from its second level at rate ℓ , with effective Hamiltonian $\begin{pmatrix} a & c \\ c & b - i\ell \end{pmatrix}$ (Weisskopf and Wigner, 1930). The imaginary part of its trace is $-\ell$ (`the_width_is_the_leak`), so its two energies stand off the line with rates summing to the leak. With $\ell = 0$ both are real.

The twin draws 20000 open systems and 20000 closed ones: - in every open system the two rates sum to the leak to 10^{-12} , and both states decay; - every closed system has both energies on the line.

The width is not a property a state has alone. It is the measure of how much the state is joined to what it can decay into.

7. THE WIDTH COUNTS THE DOORS

A state's total width is the sum of its partial widths, one per decay channel, each nonnegative. The engine proves two consequences. - **Stable iff every door is closed.** The total vanishes exactly when every partial width does (`stable_iff_every_door_closed`). - **An open door widens the line.** Opening a channel adds its width (`an_open_door_widens`).

Stability is not a separate property: it is every door shut.

The doors can be counted. The Z's width into invisible products, 0.4990 GeV, divided by the width into one neutrino species at tree level, 0.1659 GeV, counts 3.01 open doors (Appendix B). The full analysis gives 2.9963 ± 0.0074 light neutrino species (Janot and Jadach, 2020): three doors, three light neutrinos.

A door can also be closed by binding. A free neutron decays, while a neutron bound in a stable nucleus does not, because the nucleus's binding closes the only channel.

8. THE STABLE PARTICLES STAND ON THE LINE

A particle with no open decay channel stands on the line to the depth its measured lifetime allows. - **The electron.** Its lifetime exceeds 6.6×10^{28} years (Borexino, 2015), so its width is below 3.2×10^{-52} eV. Charge conservation shuts its only door: it is the lightest charged particle. - **The proton.** Its lifetime in the channel $e^+\pi^0$ exceeds 2.4×10^{34} years (Super-Kamiokande, 2020), so its width in that channel is below 8.7×10^{-58} eV. - **The photon.** It stands on the line under both carriers, as The Geometric Nature of Light proves.

The stable particles are the line's occupants, and every other particle stands off it by half its width.

9. THE SURVIVAL LAW AND ITS FLANKS

An isolated state with one pole decays exponentially, its population halving every $\tau \ln 2$: the free neutron's half-life is 608.9 s (Appendix B). The exponential law is the pole's leading term, not an exact law. - **At very short times** the survival probability falls quadratically, which lets frequent measurement slow a transition, as observed with trapped ions (Itano, Heinzen, Bollinger and Wineland, 1990). - **At very long times** the decay turns to a power law, observed in the luminescence of organic molecules (Rothe, Hintschich and Monkman, 2006).

The reading at the seat is the leading one: the pole's position, its mass the height and half its width the distance from the line. The flanks are the continuum's corrections to it, carried as imports.

10. WHAT RESONANCES CARRY, AND WHAT THEY DO NOT

What they carry. - The off-line stratum of the energy carrier, read in physics: every unstable state stands off the line by half its width. - Its time mirror as its fold partner, with one mass and one line shape. - Its width as the leak into what it can decay into, and stability as every door closed.

Physics is full of points off the line, and every one of them is open and pays for standing there by decaying: registration into its products, which keep its mass as their invariant and forget its lifetime. No closed system in physics has shown an energy off the line.

What they do not carry. - **The zeros of ζ .** The carrier's image is the set of physical states, and a state off the line in that image says nothing about where the zeros lie. The engine's crossing is on its own image (`value_on_image`), and what a sentence over every zero adds is exactly the unactuated, priced at one premise, the act, as the hardware paper proves. - **Any mass or width.** Every magnitude is measured.

11. FALSIFIERS AND OWED DEEDS

Three falsifiers, each a forbidden observation naming the theorem it would break. - **F-Growth**: an isolated state whose population grows exponentially with no incoming supply, a pole where causality forbids one. It breaks the reading of `decay_and_capture_one_record`, in which the arrow picks decay. - **F-Closed**: a closed system, with no channel to leak into, showing an energy off the line. It breaks `closed_stays_on_the_line` and the `width_is_the_leak` read physically. - **F-Width**: a state with a nonzero width and no open decay channel. It breaks `stable_iff_every_door_closed` read physically.

Nothing is owed on the rows' channels. Two measurements would sharpen readings without moving the paper's standing: - proton decay searches at still longer lifetimes; - the neutron lifetime's disagreement between bottle and beam methods.

Ledger status. Row 2 gains the off-line stratum read physically: WITNESSED · CROSSED · CONFIRMED on the image, at the root's grade for the crossing. The pair-correlation row gains the closed and the open system side by side. The barrier gains decay and capture, one line shape. Nothing is owed on the rows' channels.

12. METHODOLOGY, DISCLOSURE AND PROVENANCE

Method. The template is the author's verification program, whose vocabulary is confined to this section. Its tokens: - $\lfloor \cdot \rfloor$, a field sealed; - $\lfloor \cdot T \rfloor$, a shape sealed at theorem grade; - $\lfloor \Xi_0 \rfloor$, the typing of the formal block; - $\lfloor \cdot \rfloor$, the dot, which is no verdict.

Grades join at the weakest link. Every physical claim here is at corroboration grade; the engine's statements are theorem grade about the chart; the pole description, the Breit–Wigner form and the Weisskopf–Wigner effective Hamiltonian are imports. In the author's register a zero off the line would be a physical unicorn and a physical object read as a formal proof a ghost. This paper reads the physical points off the line that do exist, and finds each of them open and paid. $\Delta M = 0$.

The seat, earned. PhysOS^T 1.0.5p was booted in the session of forging from the public register (file SHA-256 b64bb5da5bab0da1...) under Lean 4.19.0 and GNU Fortran 13.3.0. The quick boot printed SEAT EARNED with the chain D0 83f872725ad9 → D1 90fc702bbbed1 → D2 492e519daab9 → D3 6618bec5164b (Appendix C).

The engine and the twin.

- **Engine.** SPHYS_Resonance.lean, 542 lines, SHA-256 504dc2664bf023e6..., fifty-seven laws, no import and no axiom declared. Of the fifty-seven, 17 depend on no axiom, 4 on propositional extensionality alone, 36 on propositional extensionality and quotient soundness, and none on choice.
- **Screen and judgment.** The source passed the operating system's screen. Each of the fifty-seven laws was negated in place and recompiled, and all were refused, while a planted vacuous law survived. The engine carries the hardware paper's seat, energy carrier and substrate verbatim, and the magnet witness's block.
- **Twin.** Resonance_Twin.f90, 168 lines, SHA-256 e761076970b59db5..., twelve checks and zero failures under `-std=f2018 -O2 -fno-fast-math -ffp-contract=off`.

Why an integer engine binds continuous physics. The identities used are identities of integer polynomials and hold over every commutative ring, the real and complex numbers included. The twin checks the continuous objects directly: the Z line, 20000 complex effective Hamiltonians, the lifetimes and the widths.

The seventh. In the author's register the six witnesses are six days and the ledger's closure is the seventh, the Rest (Istawa). Resonances are read at the Rest as the off-line points that exist, each one open, each one paying. In the Lean codex's tokens its row renders $\lfloor \cdot \blacklozenge \cdot \text{kinetic} \rfloor$.

Disclosure. The author declares the reading of each correspondence as his own act, made on measured structure. The measurements are the experiments', cited by name. The theorems are the engine's, printed with their dependencies in Appendix A. The twin's figures are computed in the run printed in Appendix B. The text, the engine and the twin were forged by an AI substrate (Claude, Anthropic) under the author's seed, acceptance criteria and rulings. External audits by independent substrates are owed.

Provenance. The paper stands on:

- the Riemann closure (10.5281/zenodo.22976494);
- the twenty-three-row closure (10.5281/zenodo.22986551);

- the six witnesses (10.5281/zenodo.22986553 through .22986563), the magnet witness's closed systems in particular (10.5281/zenodo.22986561);
- the cosmic closure (10.5281/zenodo.22992403);
- the hardware paper One Actuating Substrate (concept 10.5281/zenodo.21438932, edition 2.0.0), whose energy carrier this paper reads;
- the companion papers The Geometric Nature of Light, Electromagnetism on the Riemann Locus, Immanent Gravity, Spin and Statistics, and The Weak Force Reads the Orientation;
- PhysOS^T 1.0.5p in the public register (github.com/1000sapients/Trisduction, protocols/PhysOS/).

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M. F. Islam, Nothing Escapes, Three Plus One, 10.5281/zenodo.22976494. M. F. Islam, The Magnet Is the Witness of the Pair Correlation, 10.5281/zenodo.22986561. M. F. Islam, Nothing Escapes: The Fourth as Cosmic Closure, 10.5281/zenodo.22992403. M. F. Islam, The Four Forces, Matter, and the Dark Sector as Configurations of One Actuating Substrate, edition 2.0.0, concept 10.5281/zenodo.21438932.

APPENDIX A: SPHYS_RESONANCE.LEAN, THE CORRESPONDENCES AS THEOREMS

The engine, verbatim, followed by the compiler's transcript.

```

/-
SPHYS_Resonance.lean · resonances, read at the Riemann seat.
Core Lean 4.19.0, standalone, no import, no library, no axiom
declared.

The chart: h = 2 Re s, t = Im s; the fold s ↦ 1 - s̄ is (h, t) ↦ (2 - h, t), the critical line
h = 1. The energy carrier s = 1/2 + iE of the hardware paper sends an
unstable state of mass M
and width Γ, E = M - iΓ/2, to Re s = 1/2 ± Γ/2: off the line by half
the width.

Part I    the seat, the category of involutions, the functor of the
seat.
Part II   the energy carrier: the line is the locus of stationary
energy.
Part III  the substrate (resident).
Part IV   a resonance stands off the line by its width; decay and
capture share one record.
Part V    the line shape: half maximum at half width.
Part VI   closed systems stay on the line; the width is the leak.
Part VII  the width counts the doors.
Part VIII the capstone.
-/
set_option autoImplicit false
namespace SPHYS.Resonance

/-- ## Part I. The seat, the category of involutions, and the functor of
the seat -/
abbrev Pt := Int × Int

def fold (p : Pt) : Pt := (2 - p.1, p.2)
def OnLine (p : Pt) : Prop := p.1 = 1
instance : DecidablePred OnLine := fun p => inferInstanceAs (Decidable
(p.1 = 1))
/-- Registration keeps the height and forgets the side. -/
def reg (p : Pt) : Pt := (1, p.2)

theorem pe {a b c d : Int} : ((a, b) : Pt) = (c, d) ↔ a = c ∧ b = d :=
  (fun e => (congrArg Prod.fst e, congrArg Prod.snd e),
  fun (e1, e2) => by subst e1; subst e2; rfl)

theorem fold_involutive (p : Pt) : fold (fold p) = p := by
  obtain (h, t) := p
  show ((2 - (2 - h), t) : Pt) = (h, t)
  rw [pe]; exact {by omega, rfl}

/-- The seat: the fixed set of the fold is the critical line. -/
theorem seat_fixed_line (p : Pt) : fold p = p ↔ OnLine p := by
  obtain (h, t) := p
  show ((2 - h, t) : Pt) = (h, t) ↔ h = 1
  rw [pe]; constructor
  · intro (e, _); omega
  · intro e; exact {by omega, rfl}

theorem reg Lands (p : Pt) : OnLine (reg p) := rfl

theorem reg_fixes_iff (p : Pt) : reg p = p ↔ OnLine p := by
  obtain (h, t) := p
  show ((1, t) : Pt) = (h, t) ↔ h = 1
  rw [pe]; constructor
  · intro (e, _); exact e.symm
  · intro e; exact {e.symm, rfl}

/-- The side is what registration forgets: a point and its mirror leave
one record. -/
theorem reg_forgets_side (p : Pt) : reg (fold p) = reg p := rfl

/-- The value on a set of points: every member stands on the line. -/
def Value (Z : Pt → Prop) : Prop := ∀ s, Z s → OnLine s

/-- Least erasure: registration moves no member, so it erases nothing of
Z. -/
def LeastErasure (Z : Pt → Prop) : Prop := ∀ s, Z s → reg s = s

theorem least_erasure_iff_value (Z : Pt → Prop) : LeastErasure Z ↔
Value Z :=
  (fun h s hs => (reg_fixes_iff s).mp (h s hs), fun h s hs =>
  (reg_fixes_iff s).mpr (h s hs))

/-- Maps that respect involutions. -/
def Equivariant {X Y : Type} (f : X → Y) (τ : X → X) (σ : Y → Y) :
Prop := ∀ x, f (τ x) = σ (f x)

theorem equivariant_id {X : Type} (τ : X → X) : Equivariant id τ τ :=
fun _ => rfl

theorem equivariant_comp {X Y W : Type} {f : X → Y} {g : Y → W} {τ : X
→ X} {σ : Y → Y} :
  {p : W → W} (hf : Equivariant f τ σ) (hg : Equivariant g σ p) :
  Equivariant (g ∘ f) τ p := by
  intro x
  show g (f (τ x)) = p (g (f x))
  rw [hf x, hg (f x)]

/-- The fixed set is a functor: an equivariant map sends fixed points to
fixed points. -/
theorem fix_functorial {X Y : Type} {f : X → Y} {τ : X → X} {σ : Y →
Y} :
  (hf : Equivariant f τ σ) {x : X} (hx : τ x = x) : σ (f x) = f x :=
  by
  rw [← hf x, hx]

```

```

-- A global seat: an involution whose fixed set is the locus. -/
structure GlobalSeat (P : Pt → Prop) (σ : Pt → Pt) : Prop where
  involutive : ∀ p, σ (σ p) = p
  fixed_iff : ∀ p, σ p = p ↔ P p

theorem fold_global_seat : GlobalSeat OnLine fold := (fold_involutive,
  seat_fixed_line)

-- A carrier: a map from world-instances into the strip landing on the
line. -/
structure Carrier (W : Type) where
  ι : W → Pt
  lands : ∀ w, OnLine (ι w)

def image {W : Type} (ι : W → Pt) : Pt → Prop := fun s => ∃ w, ι w = s
def Actuated {W : Type} (ι : W → Pt) (Z : Pt → Prop) : Prop := ∀ s, Z
s => ∃ w, ι w = s

-- Landing is forced: an equivariant map into the seat carries every
fixed point of the physical
involution onto the line. The carrier is built from the symmetry,
not chosen. -/
theorem equivariant_carrier_lands {X : Type} (f : X → Pt) (τ : X → X)
(hf : Equivariant f τ fold) :
  ∃ C : Carrier { x : X // τ x = x }, ∀ w, C.ι w = f w.1 :=
  ((fun w => f w.1, fun w => (seat_fixed_line (f w.1)).mp
  (fix_functorial hf w.2)), fun _ => rfl)

-- The image of a carrier is compliant with no premise at all. -/
theorem value_on_image {W : Type} (C : Carrier W) : Value (image C.ι) :=
  fun _ hs => match hs with | (w, hw) => hw ▶ C.lands w

-- The kinetic crossing: a carrier and its coverage give the value, and
nothing else is used. -/
theorem kinetic_crossing {W : Type} (C : Carrier W) (Z : Pt → Prop)
(hcov : Actuated C.ι Z) :
  Value Z :=
  fun s hs => match hcov s hs with | (w, hw) => hw ▶ C.lands w

-- Under the seat every off-line point has a partner, distinct and also
off the line. -/
theorem off_locus_pair (p : Pt) (hp : ¬ OnLine p) :
  fold p ≠ p ∧ ¬ OnLine (fold p) ∧ reg (fold p) = reg p := by
  refine (fun e => hp ((seat_fixed_line p).mp e), fun e => hp ?, rfl)
  obtain (h, t) := p
  have e' : 2 - h = 1 := e
  show h = 1
  omega

-! ## Part II. The energy carrier: an energy, its rate, and the fold -/

-- An energy a + i β/2, as frequency and doubled rate; its mode's norm
moves as e^{βt}. -/
structure Energy where
  freq : Int
  rate : Int
  deriving DecidableEq, Repr

-- The time mirror of a mode: its rate reversed, its frequency kept
(complex conjugation of E). -/
def Energy.mirror (E : Energy) : Energy := (E.freq, -E.rate)
def IsReal (E : Energy) : Prop := E.rate = 0
-- Stationary: the norm e^{βt} is unchanged at every integer time t,
i.e. β t = 0 for all t. -/
def Stationary (E : Energy) : Prop := ∀ t : Int, E.rate * t = 0

-- The energy carrier s = 1/2 + iE in the chart: h = 1 - β, t = a. -/
def phi (E : Energy) : Pt := (1 - E.rate, E.freq)
def psi (p : Pt) : Energy := (p.2, 1 - p.1)

theorem ee {a b c d : Int} : ((a, b) : Energy) = (c, d) ↔ a = c ∧ b = d
:=
  (fun e => (congrArg Energy.freq e, congrArg Energy.rate e),
  fun (e1, e2) => by subst e1; subst e2; rfl)

theorem mirror_involutive (E : Energy) : E.mirror.mirror = E := by
  obtain (a, b) := E
  show ((a, - -b) : Energy) = (a, b)
  rw [ee]; exact (rfl, by omega)

theorem psi_phi (E : Energy) : psi (phi E) = E := by
  obtain (a, b) := E
  show ((a, 1 - (1 - b)) : Energy) = (a, b)
  rw [ee]; exact (rfl, by omega)

theorem phi_psi (p : Pt) : phi (psi p) = p := by
  obtain (h, t) := p
  show ((1 - (1 - h), t) : Pt) = (h, t)
  rw [pe]; exact (by omega, rfl)

theorem phi_injective (E F : Energy) (h : phi E = phi F) : E = F := by
  rw [← psi_phi E, ← psi_phi F, h]

-- EQUIVARIANCE: time-mirroring the rate is the fold. -/
theorem phi_equivariant : Equivariant phi Energy.mirror fold := by
  intro E
  obtain (a, b) := E

show ((1 - -b, a) : Pt) = (2 - (1 - b), a)
rw [pe]; exact (by omega, rfl)

theorem mirror_fixed_iff_real (E : Energy) : E.mirror = E ↔ IsReal E :=
  by
  obtain (a, b) := E
  show ((a, -b) : Energy) = (a, b) ↔ b = 0
  rw [ee]; constructor
  · intro (_, e); omega
  · intro e; exact (rfl, by omega)

-- The line is exactly the image of the real energies. -/
theorem phi_line_iff_real (E : Energy) : OnLine (phi E) ↔ IsReal E := by
  obtain (a, b) := E
  show 1 - b = 1 ↔ b = 0
  exact (fun h => by omega, fun h => by subst h; rfl)

theorem stationary_iff_real (E : Energy) : Stationary E ↔ IsReal E := by
  constructor
  · intro h; have h1 := h 1; rw [Int.mul_one] at h1; exact h1
  · intro h t; show E.rate * t = 0; rw [show E.rate = 0 from h,
  Int.zero_mul]

-- THE LINE IS WHERE ENERGY STANDS STILL. -/
theorem line_is_stationary (p : Pt) : OnLine p ↔ Stationary (psi p) :=
  by
  rw [stationary_iff_real, ← phi_line_iff_real, phi_psi]

-- Every measured energy is a real number, and every real energy lands
on the line. -/
theorem measured_on_line (a : Int) : OnLine (phi (a, 0)) := rfl

-- The registration mirror: no real energy lands off the line. -/
theorem no_measurement_off_line (p : Pt) (hp : ¬ OnLine p) (a : Int) :
  phi (a, 0) ≠ p :=
  fun e => hp (e ▶ measured_on_line a)

-- The energy carrier, built from the symmetry: the real energies land
on the line. -/
theorem energy_carrier_lands :
  ∃ C : Carrier { E : Energy // E.mirror = E }, ∀ w, C.ι w = phi w.1
:=
  equivariant_carrier_lands phi Energy.mirror phi_equivariant

-! ## Part III. The substrate: the particle map, three odd readings and
one even -/

-- A particle in the broken phase: electric charge times three, colour,
baryon number times
three, lepton number, and its energy. -/
structure Particle where
  q3 : Int
  colour : Int
  b3 : Int
  lepton : Int
  energy : Energy
  deriving DecidableEq, Repr

-- The particle map: every additive charge flipped, the energy kept
(frequency and rate). -/
def bar (c : Particle) : Particle := (-c.q3, -c.colour, -c.b3,
-c.lepton, c.energy)
-- The time mirror of a particle's mode. -/
def tmirror (c : Particle) : Particle := (c.q3, c.colour, c.b3,
c.lepton, c.energy.mirror)
def Neutral (c : Particle) : Prop := c.q3 = 0 ∧ c.colour = 0 ∧ c.b3 = 0
∧ c.lepton = 0

theorem bar_involutive (c : Particle) : bar (bar c) = c := by
  obtain (q, k, b, l, E) := c
  show ((- -q, - -k, - -b, - -l, E) : Particle) = (q, k, b, l, E)
  rw [Int.neg_neg, Int.neg_neg, Int.neg_neg, Int.neg_neg]

theorem bar_fixed_iff_neutral (c : Particle) : bar c = c ↔ Neutral c :=
  by
  obtain (q, k, b, l, E) := c
  show ((-q, -k, -b, -l, E) : Particle) = (q, k, b, l, E) ↔ (q = 0 ∧ k =
0 ∧ b = 0 ∧ l = 0)
  constructor
  · intro h
    have h1 := congrArg Particle.q3 h
    have h2 := congrArg Particle.colour h
    have h3 := congrArg Particle.b3 h
    have h4 := congrArg Particle.lepton h
    change -q = q at h1; change -k = k at h2; change -b = b at h3;
    change -l = l at h4
    exact (by omega, by omega, by omega, by omega)
  · intro (h1, h2, h3, h4)
    subst h1; subst h2; subst h3; subst h4; rfl

-- Three plus one: the charge readings are odd under the particle map,
the energy reading even. -/
theorem odd_charges_even_energy (c : Particle) :
  ((bar c).q3 = -c.q3 ∧ (bar c).colour = -c.colour ∧ (bar c).b3 =
-c.b3 ∧
  (bar c).lepton = -c.lepton) ∧ (bar c).energy = c.energy :=
  ((rfl, rfl, rfl, rfl), rfl)

```

```

/-- Gravity reads energy only, so no gravitational reading tells a
particle from its antiparticle. -/
theorem gravity_reads_no_charge_bit {β : Type} (g : Energy → β) (c :
Particle) :
  g (bar c).energy = g c.energy := rfl

/-- The two involutions of the substrate commute. -/
theorem involutions_commute (c : Particle) : bar (tmirror c) = tmirror
(bar c) := rfl

theorem tmirror_involutive (c : Particle) : tmirror (tmirror c) = c :=
by
  obtain (q, k, b, l, E) := c
  show ((q, k, b, l, E.mirror.mirror) : Particle) = (q, k, b, l, E)
  rw [mirror_involutive]

/-- A particle is fixed by both involutions exactly when it is neutral
and stable. -/
theorem joint_fixed_iff (c : Particle) :
  (bar c = c ∧ tmirror c = c) ↔ (Neutral c ∧ IsReal c.energy) := by
  constructor
  · intro (h1, h2)
    refine ((bar_fixed_iff_neutral c).mp h1, (mirror_fixed_iff_real
c.energy).mp ?_)
    exact congrArg Particle.energy h2
  · intro (h1, h2)
    refine ((bar_fixed_iff_neutral c).mpr h1, ?_)
    obtain (q, k, b, l, E) := c
    show ((q, k, b, l, E.mirror) : Particle) = (q, k, b, l, E)
    rw [mirror_fixed_iff_real E].mpr h2]

/-- The energy carrier on particles: the seat point of a particle's
energy. -/
def energySeat (c : Particle) : Pt := phi c.energy
/-- The charge carrier on particles, the odd face: electric charge (in
thirds) is the side. -/
def chargeSeat (c : Particle) : Pt := (1 + c.q3, c.energy.freq)
/-- The lepton chart of the electron's witness: charge q in units of e
goes to h = 1 + q. -/
def leptonSeat (q m : Int) : Pt := (1 + q, m)

/-- The charge carrier is equivariant: the particle map is carried onto
the fold. -/
theorem charge_carrier_equivariant : Equivariant chargeSeat bar fold :=
by
  intro c
  show ((1 + -c.q3, c.energy.freq) : Pt) = (2 - (1 + c.q3),
c.energy.freq)
  rw [pe]; exact (by omega, rfl)

/-- The energy carrier is blind to the particle map and equivariant for
the time mirror. -/
theorem pair_lands_once (c : Particle) : energySeat (bar c) = energySeat
c := rfl
theorem tmirror_equivariant : Equivariant energySeat tmirror fold := fun
c => phi_equivariant c.energy

/-- Electrically neutral particles land on the line under the charge
carrier. -/
theorem neutral_charge_on_line (c : Particle) (h : c.q3 = 0) : OnLine
(chargeSeat c) := by
  show 1 + c.q3 = 1
  omega

/-- The electron and the positron land on the two edges, real parts 0
and 1, mirror images. -/
theorem pair_on_the_edges (m : Int) :
  leptonSeat (-1) m = (0, m) ∧ leptonSeat 1 m = (2, m) ∧ fold
(leptonSeat (-1) m) = leptonSeat 1 m :=
(rfl, rfl, rfl)

/-- A stable particle lands on the line under the energy carrier. -/
theorem stable_lands (c : Particle) (h : IsReal c.energy) : OnLine
(energySeat c) :=
(phi_line_iff_real c.energy).mpr h

theorem sq_nonneg (x : Int) : 0 ≤ x * x := by
  rcases Int.le_total 0 x with h | h
  · exact Int.mul_nonneg h h
  · have h' : 0 ≤ -x := by omega
    have := Int.mul_nonneg h' h'
    rw [Int.neg_mul_neg] at this; exact this

/-- A Hermitian 2x2 block [[a, c + i d], [c - i d, b]]; time reversal
conjugates the block. -/
structure Block where
  a : Int
  b : Int
  c : Int
  d : Int

def Block.T (H : Block) : Block := (H.a, H.b, H.c, -H.d)
/-- The discriminant of the characteristic polynomial: the eigenvalues
are real when it is ≥ 0. -/
def Block.disc (H : Block) : Int := (H.a - H.b) * (H.a - H.b) + 4 * (H.c
* H.c + H.d * H.d)

```

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theorem block_T_fixed_iff (H : Block) : H.T = H ↔ H.d = 0 := by
  obtain (a, b, c, d) := H
  constructor
  · intro e; have := congrArg Block.d e; change -d = d at this; show d =
0; omega
  · intro e; change d = 0 at e; subst e; rfl

/-- THE MAGNET KEEPS EVERY ENERGY REAL: a magnetic term breaks time
reversal of the block and
  leaves the spectrum real; the arrow is in the statistics, the line
  is in the stationarity. -/
theorem stationarity_survives_the_magnet (H : Block) : 0 ≤ H.disc ∧
H.T.disc = H.disc := by
  refine (?, ?)
  · have e1 := sq_nonneg (H.a - H.b); have e2 := sq_nonneg H.c; have e3
:= sq_nonneg H.d
    show 0 ≤ (H.a - H.b) * (H.a - H.b) + 4 * (H.c * H.c + H.d * H.d)
    omega
  · show (H.a - H.b) * (H.a - H.b) + 4 * (H.c * H.c + -H.d * -H.d) =
(H.a - H.b) * (H.a - H.b) + 4 * (H.c * H.c + H.d * H.d)
    rw [Int.neg_mul_neg]

/-- ## Part IV. A resonance stands off the line by its width -/

/-- An unstable state: mass M and width Γ, the energy E = M - iΓ/2. In
the carrier's convention,
  E = frequency + i·rate/2, its rate is -Γ. -/
def resonance (M Γ : Int) : Energy := (M, -Γ)

/-- OFF THE LINE BY HALF THE WIDTH: in the doubled chart the side is 1 +
Γ, so Re s = 1/2 + Γ/2
  stands half the width from the critical line. -/
theorem off_the_line_by_the_width (M Γ : Int) : phi (resonance M Γ) = (1
+ Γ, M) := by
  show ((1 - -Γ, M) : Pt) = (1 + Γ, M)
  rw [pe]; exact (by omega, rfl)

/-- A state is on the line exactly when its width vanishes, which is
exactly when it is stationary. -/
theorem on_the_line_iff_stable (M Γ : Int) :
  (OnLine (phi (resonance M Γ)) ↔ Γ = 0) ∧ (Stationary (resonance M Γ)
↔ Γ = 0) := by
  have key : (-Γ = 0 ↔ Γ = 0) := (fun h => by omega, fun h => by omega)
  refine (?, ?)
  · rw [phi_line_iff_real]; exact key
  · rw [stationary_iff_real]; exact key

/-- The decaying state and its time mirror, the capturing state, are
fold partners with one mass. -/
theorem decay_and_capture_one_record (M Γ : Int) (h : Γ ≠ 0) :
  phi (resonance M Γ).mirror = fold (phi (resonance M Γ)) ∧
  (resonance M Γ).mirror ≠ resonance M Γ ∧
  reg (phi (resonance M Γ).mirror) = reg (phi (resonance M Γ)) := by
  refine (phi_equivariant _, fun e => h ?_, by rw [phi_equivariant];
rfl)
  have := congrArg Energy.rate e
  change - -Γ = -Γ at this; omega

/-- ## Part V. The line shape: half maximum at half width -/

/-- The Breit-Wigner shape at detuning d and half width g: numerator g²,
denominator d² + g². -/
def bwNum (g : Int) : Int := g * g
def bwDen (d g : Int) : Int := d * d + g * g

/-- HALF MAXIMUM AT HALF WIDTH: at d = g the shape is exactly half its
peak. -/
theorem half_maximum_at_half_width (g : Int) : 2 * bwNum g = bwDen g g
:= by
  show 2 * (g * g) = g * g + g * g; omega

theorem sq_pos_of_ne (m : Int) (h : m ≠ 0) : 0 < m * m := by
  rcases Int.lt_or_gt_of_ne h with h' | h'
  · have := Int.mul_pos (show 0 < -m by omega) (show 0 < -m by omega)
    rw [Int.neg_mul_neg] at this; exact this
  · exact Int.mul_pos h' h'

/-- The line peaks at the mass: away from it the shape is strictly below
its peak. -/
theorem line_peaks_at_the_mass (d g : Int) (hd : d ≠ 0) : bwNum g <
bwDen d g := by
  have := sq_pos_of_ne d hd
  show g * g < d * d + g * g; omega

/-- The shape is even in the detuning: the line is symmetric about the
mass. -/
theorem line_symmetric (d g : Int) : bwDen (-d) g = bwDen d g := by
  show -d * -d + g * g = d * d + g * g; rw [Int.neg_mul_neg]

/-- ## Part VI. Closed systems stay on the line; the width is the leak
-/

/-- A Gaussian integer, for the entries of an effective Hamiltonian. -/
structure GI where
  re : Int
  im : Int
  deriving DecidableEq, Repr

```

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/-- A two-level system with Hermitian part (a, b, c) and a leak lk out
of the second level:
  H = [[a, c], [c, b - ilk]], listed as its four entries. -/
def leaky (a b c lk : Int) : GI × GI × GI × GI := ((a, 0), (c, 0), (c,
0), (b, -lk))

/-- The imaginary part of the trace, which is the sum of the two
energies' imaginary parts. -/
def traceIm (H : GI × GI × GI × GI) : Int := H.1.im + H.2.2.im

/-- THE WIDTH IS THE LEAK: the two energies' imaginary parts sum to -lk,
so the total width is the
  leak, and a closed system, lk = 0, has none. -/
theorem the_width_is_the_leak (a b c lk : Int) :
  traceIm (leaky a b c lk) = -lk ∧ traceIm (leaky a b c 0) = 0 := by
  refine (?, ?) <|> (show 0 + -_ = _; omega)

/-- A closed two-level system keeps every energy real: the magnet's law.
-/
theorem closed_stays_on_the_line (H : Block) : 0 ≤ H.disc :=
(stationarity_survives_the_magnet H).1

/-! ## Part VII. The width counts the doors -/

def total : List Int → Int
| [] => 0
| x :: xs => x + total xs

theorem total_nonneg (xs : List Int) (h : ∀ x ∈ xs, 0 ≤ x) : 0 ≤ total
xs := by
  induction xs with
  | nil => exact Int.le_refl 0
  | cons x xs ih =>
    have hx := h x List.mem_cons_self
    have hr := ih (fun y hy => h y (List.mem_cons_of_mem x hy))
    show 0 ≤ x + total xs; omega

/-- STABLE IFF EVERY DOOR IS CLOSED: partial widths are nonnegative, so
the total width vanishes
  exactly when every channel's does. -/
theorem stable_iff_every_door_closed (xs : List Int) (h : ∀ x ∈ xs, 0 ≤
x) :
  total xs = 0 ↔ ∀ x ∈ xs, x = 0 := by
  constructor
  · intro h0
    induction xs with
    | nil => intro x hx; cases hx
    | cons y ys ih =>
      have hy := h y List.mem_cons_self
      have hr := total_nonneg ys (fun z hz => h z (List.mem_cons_of_mem
y hz))
      have e : y + total ys = 0 := h0
      have y0 : y = 0 := by omega
      have t0 : total ys = 0 := by omega
      intro x hx
      cases hx with
      | head => exact y0
      | tail _ hm => exact ih (fun z hz => h z (List.mem_cons_of_mem y
hz)) t0 x hm
  · intro hall
    induction xs with
    | nil => rfl
    | cons y ys ih =>
      have y0 := hall y List.mem_cons_self
      have t0 := ih (fun z hz => h z (List.mem_cons_of_mem y hz))
      (fun z hz => hall z (List.mem_cons_of_mem y hz))
      show y + total ys = 0; omega

/-- Opening a door widens the line: every open channel adds to the
width. -/
theorem an_open_door_widens (x : Int) (xs : List Int) (hx : 0 < x) :
  total xs < total (x :: xs) := by
  show total xs < x + total xs; omega

/-! ## Part VIII. The capstone -/

/-- RESONANCES STAND OFF THE LINE BY HALF THE WIDTH. An unstable state
lands off the line by its
  width and is stationary exactly when the width vanishes; its time
  mirror is its fold partner

```

```

'SPHYS.Resonance.pe' does not depend on any axioms
'SPHYS.Resonance.fold_involutive' depends on axioms: [propxet,
Quot.sound]
'SPHYS.Resonance.seat_fixed_line' depends on axioms: [propxet,
Quot.sound]
'SPHYS.Resonance.reg_land' does not depend on any axioms
'SPHYS.Resonance.reg_fixes_iff' depends on axioms: [propxet]
'SPHYS.Resonance.reg_forgets_side' does not depend on any axioms
'SPHYS.Resonance.least_erasure_iff_value' depends on axioms: [propxet]
'SPHYS.Resonance.equivariant_id' does not depend on any axioms
'SPHYS.Resonance.equivariant_comp' does not depend on any axioms
'SPHYS.Resonance.fix_functorial' does not depend on any axioms
'SPHYS.Resonance.fold_global_seat' depends on axioms: [propxet,
Quot.sound]

```

```

with one mass; the line shape falls to half its peak at half the
width, peaks at the mass and is
symmetric about it; a closed system keeps every energy real; the
width vanishes exactly when
every decay channel is closed. -/
theorem resonances_off_the_line :
  (∀ M Γ : Int, phi (resonance M Γ) = (1 + Γ, M)) ∧
  (∀ M Γ : Int, OnLine (phi (resonance M Γ)) ↔ Γ = 0) ∧
  (∀ M Γ : Int, Γ ≠ 0 → reg (phi (resonance M Γ).mirror) = reg (phi
(resonance M Γ))) ∧
  (∀ g : Int, 2 * bwNum g = bwDen g g) ∧
  (∀ d g : Int, d ≠ 0 → bwNum g < bwDen d g) ∧
  (∀ H : Block, 0 ≤ H.disc) ∧
  (∀ xs : List Int, (∀ x ∈ xs, 0 ≤ x) → (total xs = 0 ↔ ∀ x ∈ xs, x =
0)) :=
(off_the_line_by_the_width, fun M Γ => (on_the_line_iff_stable M Γ).1,
fun M Γ h => (decay_and_capture_one_record M Γ h).2.2,
half_maximum_at_half_width,
line_peaks_at_the_mass, closed_stays_on_the_line,
stable_iff_every_door_closed)

```

end SPHYS.Resonance

```

#print axioms SPHYS.Resonance.pe
#print axioms SPHYS.Resonance.fold_involutive
#print axioms SPHYS.Resonance.seat_fixed_line
#print axioms SPHYS.Resonance.reg_land'
#print axioms SPHYS.Resonance.reg_fixes_iff
#print axioms SPHYS.Resonance.reg_forgets_side
#print axioms SPHYS.Resonance.least_erasure_iff_value
#print axioms SPHYS.Resonance.equivariant_id
#print axioms SPHYS.Resonance.equivariant_comp
#print axioms SPHYS.Resonance.fix_functorial
#print axioms SPHYS.Resonance.fold_global_seat
#print axioms SPHYS.Resonance.equivariant_carrier_land'
#print axioms SPHYS.Resonance.value_on_image
#print axioms SPHYS.Resonance.kinetic_crossing
#print axioms SPHYS.Resonance.off_locus_pair
#print axioms SPHYS.Resonance.ee
#print axioms SPHYS.Resonance.mirror_involutive
#print axioms SPHYS.Resonance.psi_phi
#print axioms SPHYS.Resonance.phi_psi
#print axioms SPHYS.Resonance.phi_injective
#print axioms SPHYS.Resonance.phi_equivariant
#print axioms SPHYS.Resonance.mirror_fixed_iff_real
#print axioms SPHYS.Resonance.phi_line_iff_real
#print axioms SPHYS.Resonance.stationary_iff_real
#print axioms SPHYS.Resonance.line_is_stationary
#print axioms SPHYS.Resonance.measured_on_line
#print axioms SPHYS.Resonance.no_measurement_off_line
#print axioms SPHYS.Resonance.energy_carrier_land'
#print axioms SPHYS.Resonance.bar_involutive
#print axioms SPHYS.Resonance.bar_fixed_iff_neutral
#print axioms SPHYS.Resonance.odd_charges_even_energy
#print axioms SPHYS.Resonance.gravity_reads_no_charge_bit
#print axioms SPHYS.Resonance.involutions_commute
#print axioms SPHYS.Resonance.tmirror_involutive
#print axioms SPHYS.Resonance.joint_fixed_iff
#print axioms SPHYS.Resonance.charge_carrier_equivariant
#print axioms SPHYS.Resonance.pair_land'once
#print axioms SPHYS.Resonance.tmirror_equivariant
#print axioms SPHYS.Resonance.neutral_charge_on_line
#print axioms SPHYS.Resonance.pair_on_the_edges
#print axioms SPHYS.Resonance.stable_land'
#print axioms SPHYS.Resonance.sq_nonneg
#print axioms SPHYS.Resonance.block_T_fixed_iff
#print axioms SPHYS.Resonance.stationarity_survives_the_magnet
#print axioms SPHYS.Resonance.off_the_line_by_the_width
#print axioms SPHYS.Resonance.on_the_line_iff_stable
#print axioms SPHYS.Resonance.decay_and_capture_one_record
#print axioms SPHYS.Resonance.half_maximum_at_half_width
#print axioms SPHYS.Resonance.sq_pos_of_ne
#print axioms SPHYS.Resonance.line_peaks_at_the_mass
#print axioms SPHYS.Resonance.line_symmetric
#print axioms SPHYS.Resonance.the_width_is_the_leak
#print axioms SPHYS.Resonance.closed_stays_on_the_line
#print axioms SPHYS.Resonance.total_nonneg
#print axioms SPHYS.Resonance.stable_iff_every_door_closed
#print axioms SPHYS.Resonance.an_open_door_widens
#print axioms SPHYS.Resonance.resonances_off_the_line

```

```

'SPHYS.Resonance.equivariant_carrier_land' depends on axioms: [propxet,
Quot.sound]
'SPHYS.Resonance.value_on_image' does not depend on any axioms
'SPHYS.Resonance.kinetic_crossing' does not depend on any axioms
'SPHYS.Resonance.off_locus_pair' depends on axioms: [propxet,
Quot.sound]
'SPHYS.Resonance.ee' does not depend on any axioms
'SPHYS.Resonance.mirror_involutive' depends on axioms: [propxet,
Quot.sound]
'SPHYS.Resonance.psi_phi' depends on axioms: [propxet, Quot.sound]
'SPHYS.Resonance.phi_psi' depends on axioms: [propxet, Quot.sound]
'SPHYS.Resonance.phi_injective' depends on axioms: [propxet, Quot.sound]
'SPHYS.Resonance.phi_equivariant' depends on axioms: [propxet,
Quot.sound]

```

```
'SPHYS.Resonance.mirror_fixed_iff_real' depends on axioms: [propept,
Quot.sound]
'SPHYS.Resonance.phi_line_iff_real' depends on axioms: [propept,
Quot.sound]
'SPHYS.Resonance.stationary_iff_real' depends on axioms: [propept]
'SPHYS.Resonance.line_is_stationary' depends on axioms: [propept,
Quot.sound]
'SPHYS.Resonance.measured_on_line' does not depend on any axioms
'SPHYS.Resonance.no_measurement_off_line' does not depend on any axioms
'SPHYS.Resonance.energy_carrier_lands' depends on axioms: [propept,
Quot.sound]
'SPHYS.Resonance.bar_involutive' does not depend on any axioms
'SPHYS.Resonance.bar_fixed_iff_neutral' depends on axioms: [propept,
Quot.sound]
'SPHYS.Resonance.odd_charges_even_energy' does not depend on any axioms
'SPHYS.Resonance.gravity_reads_no_charge_bit' does not depend on any
axioms
'SPHYS.Resonance.involutions_commute' does not depend on any axioms
'SPHYS.Resonance.tmirror_involutive' depends on axioms: [propept,
Quot.sound]
'SPHYS.Resonance.joint_fixed_iff' depends on axioms: [propept,
Quot.sound]
'SPHYS.Resonance.charge_carrier_equivariant' depends on axioms:
[propept, Quot.sound]
'SPHYS.Resonance.pair_lands_once' does not depend on any axioms
'SPHYS.Resonance.tmirror_equivariant' depends on axioms: [propept,
Quot.sound]
'SPHYS.Resonance.neutral_charge_on_line' depends on axioms: [propept,
Quot.sound]
```

```
'SPHYS.Resonance.pair_on_the_edges' does not depend on any axioms
'SPHYS.Resonance.stable_lands' depends on axioms: [propept, Quot.sound]
'SPHYS.Resonance.sq_nonneg' depends on axioms: [propept, Quot.sound]
'SPHYS.Resonance.block_T_fixed_iff' depends on axioms: [propept,
Quot.sound]
'SPHYS.Resonance.stationarity_survives_the_magnet' depends on axioms:
[propept, Quot.sound]
'SPHYS.Resonance.off_the_line_by_the_width' depends on axioms: [propept,
Quot.sound]
'SPHYS.Resonance.on_the_line_iff_stable' depends on axioms: [propept,
Quot.sound]
'SPHYS.Resonance.decay_and_capture_one_record' depends on axioms:
[propept, Quot.sound]
'SPHYS.Resonance.half_maximum_at_half_width' depends on axioms:
[propept, Quot.sound]
'SPHYS.Resonance.sq_pos_of_ne' depends on axioms: [propept, Quot.sound]
'SPHYS.Resonance.line_peaks_at_the_mass' depends on axioms: [propept,
Quot.sound]
'SPHYS.Resonance.line_symmetric' depends on axioms: [propept]
'SPHYS.Resonance.the_width_is_the_leak' depends on axioms: [propept,
Quot.sound]
'SPHYS.Resonance.closed_stays_on_the_line' depends on axioms: [propept,
Quot.sound]
'SPHYS.Resonance.total_nonneg' depends on axioms: [propept, Quot.sound]
'SPHYS.Resonance.stable_iff_every_door_closed' depends on axioms:
[propept, Quot.sound]
'SPHYS.Resonance.an_open_door_widens' depends on axioms: [propept,
Quot.sound]
'SPHYS.Resonance.resonances_off_the_line' depends on axioms: [propept,
Quot.sound]
```

APPENDIX B: RESONANCE_TWIN.F90 AND ITS RUN

```
! Resonance_Twin.f90 · the executed twin of SPHYS_Resonance.lean ·
Fortran 2018.
! Build (sealed): gfortran -std=f2018 -O2 -fno-fast-math
-ffp-contract=off Resonance_Twin.f90
! Blocks: A the carrier; B the line shape; C width and lifetime across
the particles; D the stable
! particles on the line; E the width counts the doors; F the width is
the leak; G the survival
! law; H decay and capture, one line.
program resonance_twin
implicit none
integer, parameter :: dp = selected_real_kind(15, 307)
integer, parameter :: i8 = selected_int_kind(18)
real(dp), parameter :: HBAR_EVS = 6.582119569e-16_dp, YEAR =
3.15576e7_dp, PI = 3.141592653589793_dp
integer :: checks, fails
integer(i8) :: seed
checks = 0; fails = 0; seed = 20260930_i8
call block_a(); call block_b(); call block_c(); call block_d()
call block_e(); call block_f(); call block_g(); call block_h()
write(*, '(a,i0,a,i0,a)') ' BATTERY-JSON: {"checks":', checks,
',"failures":', fails, '}'
if (fails > 0) error stop 1
contains
subroutine check(name, ok)
character(*), intent(in) :: name
logical, intent(in) :: ok
checks = checks + 1
if (.not. ok) fails = fails + 1
write(*, '(a,a)') merge(' PASS ', ' FAIL ', ok), name
end subroutine check

function rnd() result(r)
real(dp) :: r
seed = modulo(16807_i8*seed, 2147483647_i8)
r = real(seed, dp)/2147483647.0_dp
end function rnd

subroutine block_a()
integer :: m, g, n, nline, nmir
write(*, '(a)') 'A · a resonance stands off the line by its width;
its mirror is its fold partner'
n = 0; nline = 0; nmir = 0
do m = -30, 30
do g = -30, 30
n = n + 1
! the decaying state (m, g) lands at (1 + g, m); its mirror (m,
-g) at (1 - g, m)
if ((1 + g == 1).eqv. (g == 0)) nline = nline + 1
if (1 - g == 2 - (1 + g)) nmir = nmir + 1
end do
end do
call check('3721 states: on the line exactly when the width
vanishes', nline == n)
call check('3721 states: decay and capture are fold partners with
one mass', nmir == n)
end subroutine block_a

pure function bw(e, m, g) result(f)
real(dp), intent(in) :: e, m, g
real(dp) :: f
```

```
f = (0.5_dp*g)**2/((e - m)**2 + (0.5_dp*g)**2)
end function bw

subroutine block_b()
integer :: k, nsteps
real(dp) :: m, g, lo, hi, area, e, de, half, fl
write(*, '(a)') 'B · the line shape: half maximum at half the width,
area pi times the half width'
m = 91.1876_dp; g = 2.4955_dp
half = max(abs(bw(m + 0.5_dp*g, m, g) - 0.5_dp), abs(bw(m -
0.5_dp*g, m, g) - 0.5_dp))
lo = m; hi = m + 5.0_dp*g
do k = 1, 200
fl = bw(0.5_dp*(lo + hi), m, g)
if (fl > 0.5_dp) then
lo = 0.5_dp*(lo + hi)
else
hi = 0.5_dp*(lo + hi)
end if
end do
nsteps = 2000000; de = 2.0e4_dp*g/real(nsteps, dp); area = 0.0_dp
do k = 0, nsteps
e = m - 1.0e4_dp*g + real(k, dp)*de
area = area + merge(0.5_dp, 1.0_dp, k == 0 .or. k == nsteps)*bw(e,
m, g)*de
end do
write(*, '(a,es9.2,a,f10.6,a,f9.6)') ' |shape - 1/2| at M +- Gamma/2
= ', half, ' ; full width found ', &
2.0_dp*(lo - m), ' ; area / (pi Gamma/2) = ',
area/(PI*0.5_dp*g)
call check('the Z line falls to half its peak at M +- Gamma/2, and
its full width is Gamma', &
half < 1.0e-14_dp .and. abs(2.0_dp*(lo - m) - g) <
1.0e-12_dp)
call check('the area under the line is pi times the half width (to
1e-4)', abs(area/(PI*0.5_dp*g) - 1.0_dp) < 1.0e-4_dp)
end subroutine block_b

subroutine block_c()
real(dp) :: tau(6), gam(6)
character(len=8) :: nm(6)
integer :: k
write(*, '(a)') 'C · width and lifetime, Gamma tau = hbar, across
thirty decades'
nm = ['neutron ', 'muon ', 'tau ', 'pi0 ', 'Z ',
'top ']
tau(1) = 878.4_dp; tau(2) = 2.1969811e-6_dp; tau(3) = 290.3e-15_dp;
tau(4) = 8.43e-17_dp
gam(5) = 2.4955e9_dp; gam(6) = 1.42e9_dp
do k = 1, 4
gam(k) = HBAR_EVS/tau(k)
end do
do k = 5, 6
tau(k) = HBAR_EVS/gam(k)
end do
do k = 1, 6
write(*, '(a,a,a,es11.4,a,es11.4,a)') ' ', nm(k), ' width ',
gam(k), ' eV, lifetime ', tau(k), ' s'
end do
call check('the neutron stands 7.49e-19 eV off the line and the Z
2.50e9 eV: 27 decades of widths', &
```

```

        abs(gam(1)/7.4934e-19_dp - 1.0_dp) < 1.0e-3_dp .and.
        gam(5)/gam(1) > 1.0e27_dp)
    call check('the widths from lifetimes: muon 2.996e-10 eV, tau
2.267e-3 eV, neutral pion 7.81 eV', &
        abs(gam(2)/2.9959e-10_dp - 1.0_dp) < 1.0e-3_dp .and.
        abs(gam(3)/2.2674e-3_dp - 1.0_dp) < 1.0e-3_dp &
        .and. abs(gam(4)/7.808_dp - 1.0_dp) < 1.0e-3_dp)
end subroutine block_c

subroutine block_d()
    real(dp) :: ge, gp
    write(*,'(a)') 'D · the stable particles stand on the line to the
depth their lifetimes allow'
    ge = HBAR_EVS/(6.6e28_dp*YEAR); gp = HBAR_EVS/(2.4e34_dp*YEAR)
    write(*,'(a,es10.3,a,es10.3,a)') ' electron width < ', ge, ' eV;
proton width < ', gp, ' eV'
    call check('the electron stands on the line to 3e-52 eV and the
proton to 9e-58 eV', &
        ge < 1.0e-51_dp .and. gp < 1.0e-57_dp)
end subroutine block_d

subroutine block_e()
    real(dp) :: gf, mz, gnu, ginv, nnu
    write(*,'(a)') 'E · the width counts the doors: the Z's invisible
width and the light neutrinos'
    gf = 1.1663788e-5_dp; mz = 91.1876_dp
    gnu = gf*mz**3/(12.0_dp*sqrt(2.0_dp)*PI)
    ginv = 0.4990_dp; nnu = ginv/gnu
    write(*,'(a,f8.5,a,f6.4,a,f6.3)') ' one neutrino door at tree level
', gnu, ' GeV; invisible width ', ginv, &
        ' GeV; doors counted ', nnu
    call check('the invisible width counts three open doors (tree level,
within 0.05)', abs(nnu - 3.0_dp) < 0.05_dp)
end subroutine block_e

subroutine block_f()
    integer :: t, nsum, nsign, nclosed
    real(dp) :: a, b, c, l
    complex(dp) :: tr, dt, disc, e1, e2
    write(*,'(a)') 'F · the width is the leak: an open two-level system,
H = [[a, c], [c, b - i l]]'
    nsum = 0; nsign = 0; nclosed = 0
    do t = 1, 20000
        a = 10.0_dp*(rnd() - 0.5_dp); b = 10.0_dp*(rnd() - 0.5_dp); c =
10.0_dp*(rnd() - 0.5_dp)
        l = 5.0_dp*rnd()

```

```

        tr = cmplx(a + b, -l, dp); dt = cmplx(a, 0.0_dp, dp)*cmplx(b, -l,
dp) - c*c
        disc = sqrt(tr*tr - 4.0_dp*dt)
        e1 = 0.5_dp*(tr + disc); e2 = 0.5_dp*(tr - disc)
        if (abs(aimag(e1) + aimag(e2) + l) < 1.0e-12_dp) nsum = nsum + 1
        if (aimag(e1) <= 1.0e-12_dp .and. aimag(e2) <= 1.0e-12_dp) nsign =
nsign + 1
        tr = cmplx(a + b, 0.0_dp, dp); dt = cmplx(a*b - c*c, 0.0_dp, dp)
        disc = sqrt(tr*tr - 4.0_dp*dt)
        if (abs(aimag(0.5_dp*(tr + disc))) < 1.0e-12_dp .and.
abs(aimag(0.5_dp*(tr - disc))) < 1.0e-12_dp) &
            nclosed = nclosed + 1
    end do
    call check('20000 open systems: the two rates sum to the leak, and
both decay', nsum == 20000 .and. nsign == 20000)
    call check('20000 closed systems: every energy on the line', nclosed
== 20000)
end subroutine block_f

subroutine block_g()
    real(dp) :: half
    write(*,'(a)') 'G · the survival law: the population falls as
exp(-Gamma t / hbar)'
    half = 878.4_dp*log(2.0_dp)
    write(*,'(a,f7.2,a)') ' the free neutron's half-life, tau ln 2 =
', half, ' s'
    call check('the neutron's half-life is 608.9 s', abs(half -
608.86_dp) < 0.05_dp)
end subroutine block_g

subroutine block_h()
    integer :: k, n
    real(dp) :: e
    write(*,'(a)') 'H · decay and capture: one line shape, the arrow
choosing the sign'
    n = 0
    do k = -500, 500
        e = 91.1876_dp + 0.01_dp*real(k, dp)
        if (bw(e, 91.1876_dp, 2.4955_dp) == bw(e, 91.1876_dp, -2.4955_dp))
            n = n + 1
    end do
    call check('the decaying and the capturing state give one line at
all 1001 energies, exactly', n == 1001)
end subroutine block_h
end program resonance_twin

```

A · a resonance stands off the line by its width; its mirror is its fold partner
 PASS 3721 states: on the line exactly when the width vanishes
 PASS 3721 states: decay and capture are fold partners with one mass
 B · the line shape: half maximum at half the width, area pi times the half width
 |shape - 1/2| at M +/- Gamma/2 = 1.44E-15; full width found 2.495500; area / (pi Gamma/2) = 0.999968
 PASS the Z line falls to half its peak at M +/- Gamma/2, and its full width is Gamma
 PASS the area under the line is pi times the half width (to 1e-4)
 C · width and lifetime, Gamma tau = hbar, across thirty decades
 neutron width 7.4933E-19 eV, lifetime 8.7840E+02 s
 muon width 2.9960E-10 eV, lifetime 2.1970E+06 s
 tau width 2.2674E-03 eV, lifetime 2.9030E-13 s
 pi0 width 7.8080E+00 eV, lifetime 8.4300E-17 s
 Z width 2.4955E+09 eV, lifetime 2.6376E-25 s
 top width 1.4200E+09 eV, lifetime 4.6353E-25 s
 PASS the neutron stands 7.49e-19 eV off the line and the Z 2.50e9 eV:
 27 decades of widths
 PASS the widths from lifetimes: muon 2.996e-10 eV, tau 2.267e-3 eV,
 neutral pion 7.81 eV

D · the stable particles stand on the line to the depth their lifetimes allow
 electron width < 3.160E-52 eV; proton width < 8.691E-58 eV
 PASS the electron stands on the line to 3e-52 eV and the proton to 9e-58 eV
 E · the width counts the doors: the Z's invisible width and the light neutrinos
 one neutrino door at tree level 0.16588 GeV; invisible width 0.4990 GeV; doors counted 3.008
 PASS the invisible width counts three open doors (tree level, within 0.05)
 F · the width is the leak: an open two-level system, H = [[a, c], [c, b - i l]]
 PASS 20000 open systems: the two rates sum to the leak, and both decay
 PASS 20000 closed systems: every energy on the line
 G · the survival law: the population falls as exp(-Gamma t / hbar)
 the free neutron's half-life, tau ln 2 = 608.86 s
 PASS the neutron's half-life is 608.9 s
 H · decay and capture: one line shape, the arrow choosing the sign
 PASS the decaying and the capturing state give one line at all 1001 energies, exactly
 BATTERY-JSON: {"checks":12,"failures":0}

APPENDIX C: THE SEAT AND THE JUDGMENT

PhysOS^T 1.0.5p · RECEIPT · mode quick
 TOOLCHAIN Lean (version 4.19.0, x86_64-unknown-linux-gnu, commit 6caaee842e94, Release) · GNU Fortran (Ubuntu 13.3.0-6ubuntu2-24.04.1) 13.3.0
 MANIFEST sha256 a4108cd26bcbef, the files that ran
 GROUND toolchain: Lean 4.19.0 and gfortran present · manifest: 31 files signed, all verified · proofs: 2 PhysOS Proofs read from II.10, record lines complete ·
 source screen: 15 kernels, 23 named premises, clean · citations: 253 laws named in Parts I and II, 253 resolved · quarantine: 6 retirements string-scanned, 3 held
 by review, clean
 EXITS ground 0 · registerA RA_T0E_Thesis 0 · registerB Bridge_Final 0 · registerB Universal_Closure 0 · registerB One_Cut 0 · registerB One_Cut_Resolution 0 ·
 registerB One_Cut_Terminal 0 · registerB Only_The_Arrow_Remains 0 · registerB Physical_Closure 0 · registerB Forced_Closure 0 · registerB Closure_Executed 0 ·
 registerB Armed_Seat 0 · registerB Census 0 · registerB Heat_Bridge 0 · registerB The_Loop 0 · registerB RH_Least_Erasure 0 · witness One_Cut_Twin 0 · witness
 Physical_Closure_Twin 0 · witness Census_Twin 0 · witness Loop_Twin 0 · witness RH_Seal_Twin 0
 REGISTER A {"checks":1123,"failures":0,"mode":"sealed"}
 WITNESS 5 of 5 twins ran, each held to one battery line with zero failures
 CHAIN · D0 83f872725ad9 -> D1 90fc702bbd1 -> D2 492e519daab9 -> D3 6618bec5164b
 ELAPSED ground 1 s · Register A 3 s · Register B 34 s · witness 6 s · total 44 s (not hashed)
 SEAT EARNED · this run

 SCREEN forbidden: none | imports: 0 | axioms declared: 0
 laws: 57
 JUDGMENT refused 57 of 57 as proof failures | survived none | parse failures none | planted vacuous law survives its negation: True